EVERY ONE-DIMENSIONAL MINIMAL PRIME OF THE COMPLETION OF A NOETHERIAN LOCAL RING IS THE RADICAL OF THE EXTENSION OF A MINIMAL PRIME OF THE HENSELIZATION

THE STACKS PROJECT AUTHORS

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Editorial note. The proof is carried out with the auxiliary prime p = q cap A introduced only near the end; that order is the author's and is reproduced. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Henselization and completion of a Noetherian local ring, and their agreement. used at proof lines 32109-32110. Is the reduction that lets A be assumed henselian.
- The quotient-by-idempotent lemma for a quotient of an adic completion killing a square-zero annihilated ideal. invoked at proof lines 32111-32120. Produces the finite type A-algebra C whose completion is A-hat/J.
- The structure of unramified ring maps as quotients of etale algebras, and the rigidity of etale algebras over a henselian local ring with trivial residue extension. used at proof lines 32126-32133. Forces B = A after localization and hence C = A/I.
- Exactness and flatness of the completion map. used at proof lines 32134-32137. Gives J = IA-hat and I = J cap A, which turn the containment into the radical statement.

Source reference. Original lemma 02FM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3197–3197. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 817–817. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0395: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4260–4260. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 10979: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10979–10979. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32037: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32037–32037. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a Noetherian local ring with henselization A^h. Let $\mathfrak{q} \subset A^\wedge$ be a minimal prime with $\dim(A^\wedge/\mathfrak{q}) = 1$. Then there exists a minimal prime $\mathfrak{q}^h$ of A^h such that $\mathfrak{q} = \sqrt{\mathfrak{q}^h A^\wedge}$.*

Proof. Since the completion of A and A^h are the same, we may assume that A is henselian (Lemma 10979). We will apply Lemma 32037 to $A^\wedge \to A^\wedge/J$ where $J = \mathrm{Ker}(A^\wedge \to (A^\wedge)_\mathfrak{q})$. Since $\dim((A^\wedge)_\mathfrak{q}) = 0$ we see that $\mathfrak{q}^n \subset J$ for some n. Hence J/J^2 is annihilated by $\mathfrak{q}^n$. On the other hand $(J/J^2)_\mathfrak{q} = 0$ because $J_\mathfrak{q} = 0$. Hence $\mathfrak{m}$ is the only associated prime of J/J^2 and we find that a power of $\mathfrak{m}$ annihilates J/J^2. Thus the lemma applies and we find that $A^\wedge/J = C^\wedge$ for some finite type A-algebra C.

Then $C/\mathfrak{m}C = A/\mathfrak{m}$ because $A^\wedge/J$ has the same property. Hence $\mathfrak{m}_C = \mathfrak{m}C$ is a maximal ideal and $A \to C$ is unramified at $\mathfrak{m}_C$ (Algebra, Lemma 02FM). After replacing C by a principal localization we may assume that C is a quotient of an étale A-algebra B, see Algebra, Proposition 0395. However, since the residue field extension of $A \to C_{\mathfrak{m}_C}$ is trivial and A is henselian, we conclude that $B = A$ again after a localization. Thus $C = A/I$ for some ideal $I \subset A$ and it follows that $J = IA^\wedge$ (because completion is exact in our situation by Algebra, Lemma 00MB) and $I = J \cap A$ (by flatness of $A \to A^\wedge$). Since $\mathfrak{q}^n \subset J \subset \mathfrak{q}$ we see that $\mathfrak{p} = \mathfrak{q} \cap A$ satisfies $\mathfrak{p}^n \subset I \subset \mathfrak{p}$. Then $\sqrt{\mathfrak{p}A^\wedge} = \mathfrak{q}$ and the proof is complete. □